
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its
2 refutation by overlapping authors; this paper joins them and locates the edge
3 of the set that refutation relies on. Conformal PID states long-run coverage for
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the
5 deployed threshold inside an admissible radius. The claim was acted on; the
6 readout’s placement and the corollary’s derivation are conceded by name. At
7 the null scorecaster that condition is *tight*. A legal dead band on the *completed*
8 threshold leaves that set at its radius, and past it miscoverage is 1.000000 against
9 $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge belongs to the
10 saturator–scorecaster pair, not the readout; elsewhere the condition is sufficient
11 only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers; at $\hat{q} \equiv -b/2$ both
12 it and partial adjustment at $w = 0.999$ miscover.

13 1 Introduction

14 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
15 by the same first and last authors, proves in the same notation that a bounded perturbation of the
16 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the
17 joining is what this paper adds.

18 An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round, and
19 that movement is not an unnamed quantity. Forecast stability names, scores and prices it [Tunc et al.,
20 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al.,
21 2026, Van Belle et al., 2026]. Holding the level fixed and comparing the temporal structure came
22 earlier still, in forecast verification, which matches predictive marginals on real producers [Gneiting
23 et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. A third strand reports that the two
24 numbers usually printed of a deployed interval, coverage and mean length, under-describe it [Min
25 et al., 2026, Vaze, 2026]. None of it is claimed here.

26 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-
27 gelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that
28 it be predictable and lie in $[-b/2, b/2]$: “any function of the past” (Theorem 1). Such a readout
29 is already quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks
30 the theoretical coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines
31 paragraph, verbatim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025,
32 Corollary 2] add any predictable d_t , $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed
33 threshold; since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result “has exactly
34 the CPID error-integration form” and Proposition 2’s own finite-horizon bound on $|E_T|$ is retained,
35 whence $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s long-run coverage [Wang
36 and Hyndman, 2024]. Four surfaces were searched for both identifiers and “scorecaster”. They are
37 arXiv full text, arXiv abstracts, OpenReview, and the six works Semantic Scholar records citing

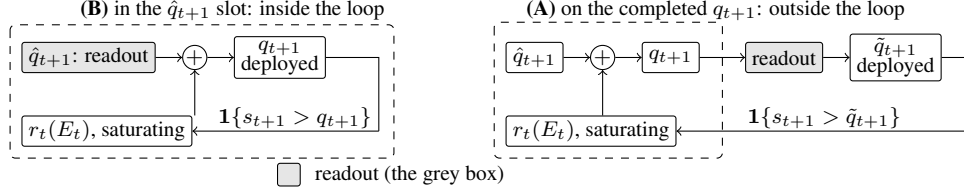


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

38 Li and Rodríguez [2025]. None returns both the claim and the corollary. Hu et al. [2026] still call
 39 scorecaster choice “arbitrary” and lacking “principled guidance” while asserting valid coverage:
 40 model selection, not validity. **We claim neither the placement nor the derivation.**

41 **Where Corollary 2 stops, its radius condition is tight.** It is a *retention* result, silent outside
 42 that radius: perturbations inside the radius keep the guarantee, and perturbations outside it are not
 43 addressed either way. A sufficient condition can be conservative, so the question the corollary leaves
 44 open is whether its radius is sharp, that is, whether something legal just outside it actually fails.
 45 Section 3 exhibits a legal perturbation leaving it: the L_1 dead-band family characterised here on the
 46 *completed* threshold, whose edge coincides with the corollary’s radius at the null scorecaster. Past
 47 it coverage fails outright, not merely its rate: the published sufficient condition is necessary there.
 48 Section 4 sets both claims beside four vocabularies for the same movement; Appendix B reads E_t as
 49 a capital process and builds a calibration test on it.

50 2 Setup

51 **The producer.** At time t a forecaster fit on data before t induces a score s_t , and a threshold
 52 q_t is emitted before y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t =$
 53 $\mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model
 54 on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score
 55 scale, generalizing the running-error feedback that adaptive conformal inference [Gibbs and Candès,
 56 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already apply directly to the
 57 threshold:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

58 Angelopoulos et al.’s iteration (5). Its r_t saturates: condition (4) requires $r_t(x) \geq b$ for $x \geq c h(t)$
 59 and $r_t(x) \leq -b$ for $x \leq -c h(t)$, with h nonnegative, nondecreasing and sublinear. In words: once
 60 the accumulated error clears $c h(t)$ in either direction, the correction is at least a full b in the direction
 61 that reduces it. With $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$, Proposition 2 gives $|E_T| \leq c h(T) + 1$, read on realised
 62 scores; that is a coverage gap of at most $(c h(T) + 1)/T$, which vanishes because h is sublinear.

63 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard
 64 readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}_{t+1}^{\text{raw}} + \lambda\tilde{q}_t$
 65 [Gârleanu and Pedersen, 2013], which Godahewa et al. [2025] publish as post-processing and Binny
 66 and Dixit [2025, Eq. 13] apply to a deployed conformal threshold. A proportional cost gives a
 67 no-trade band, moving only when the target is more than τ away, then pulling back exactly τ
 68 [Constantinides, 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in
 69 Figure 1. **(A)**, Placement A, is the one measured here, and it is the one a designer reaches for, because
 70 the deployed number is what has to sit still. *The dead band (L_1) is not a stylistic alternative to the*
 71 *partial-adjustment (L_2) form*; it is the family whose failure Section 3 characterises. Neither is safe by
 72 construction: L_2 forfeits Proposition 2’s rate while covering, L_1 past the radius loses coverage, and
 73 at the legal $\hat{q} \equiv -b/2$ both lose it. So the tight quantity is the radius, not either form.

74 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
 75 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new, and the name is
 76 a convenience rather than a contribution. Over a predictive distribution $\sum |\Delta q|$ is the Multi-Quantile
 77 Change [Zanotti, 2026, v3, §3.4.1]; verification calls the same movement *jumpiness* [Zsótér et al.,

2009], a *convergence index* [Ehret, 2010] and forecast *(in)consistency* [Pappenberger et al., 2011], all three imported by Van Belle et al. [2026]; applied control calls it IACER [Sarhadi, 2025]; and dynamic-regret online learning calls it a path length. Section 4 names ten across four vocabularies, so a fresh one is used here, for one reading only.

Harness. The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$, with $c = 1$ and $h(t) = \log(t + 2)$, meets (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated variable. Neither choice is neutral, and each sits at an edge of what condition (4) and Theorem 1 allow rather than in the middle of it: (4) is a lower bound, and this saturator meets it exactly rather than exceeding it, while Theorem 1 permits scorecasters anywhere within $b/2$ either side of null, predictable but not required to be constant in t . The adversary is specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”, firing the strict indicator. Legality is Theorem 1’s $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight. *That specification names one member of a class*: any adapted $s_t \in [-b/2, b/2]$ is legal, the constant $s_t \equiv -b/2$ included. Every number in Table 1 is measured against the one adversary above; Section 3’s two boundaries are quantified over the class, and they differ.

3 The edge of the admissible set, located and proved

The retention result is published, and tested from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \leq c h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight w leaves it; Table 1 prices that departure, and five of its eleven arms exceed the bound at $T = 10^6$, and for $w = 0.99$ and 0.999 the excursion is *flat in* T while the bound grows like $\log T$. The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311 to 202. The forfeit’s sign is counter-intuitive, so it is worth saying what does and does not break. A readout on the completed q_{t+1} touches neither r_t nor the integrator’s argument, so it cannot put (4) at risk. What it does instead is delay a correction that has already been computed, so the accumulator excurses *further* rather than less: smoothing the output does not damp E_t but feeds it.

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every number is the *primary* regime’s, defined in Appendix A, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t 10^6$	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

The L_1 dead band leaves the radius, and the edge is a theorem. A dead band pulls the deployed threshold back by exactly τ when it releases and holds it otherwise, so $|\hat{q}_{t+1} - q_{t+1}| \leq \tau$ at every round and “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator’s ceiling, and the deployed value’s realised supremum is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. Outside the radius the answer turns on two different levels of the integrator, and condition (4) names only one of them. The two are the largest correction r_t ever issues and the correction it is guaranteed to issue once the accumulator has cleared the radius; where they differ, a boundary written with the first of them over-certifies. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for the integrator’s extremes, and $\Lambda_t^+ = \inf_{x \geq c h(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \leq -c h(t)} r_t(x)$ for the levels it is *guaranteed* to reach once the accumulator clears the radius. Condition (4) says

116 exactly $\Lambda_t^+ \geq b$ and $\Lambda_t^- \leq -b$; that $\Lambda_t^\pm = A_t^\pm$, that the integrator reaches its extremes where (4) puts
 117 them, is a further hypothesis and not one (4) supplies. Set

$$\tau^{*\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

118 **Failure, over the whole admissible class.** If $\tau > \tau^{*+}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round,
 119 so $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a
 120 hold or a release downward cannot lift the deployed value at all. Against $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the
 121 indicator is then $\mathbf{1}\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \geq \tau^{*-}$ the constant legal play $s_t \equiv -b/2$
 122 makes $\{\tilde{q} \geq -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate
 123 is 0 rather than α . Either way long-run coverage is lost, with $|E_T|$ linear in T . Nothing here asks $\hat{q}$
 124 to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the
 125 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \leq \sigma^+$
 126 then $E_{t-1} \geq c h(t-1)$ forces $q_t \geq b/2 + \tau$ and so $\tilde{q}_t \geq b/2$, whence $\text{err}_t = 0$ against every legal
 127 adversary; if $\tau < \sigma^-$ the mirror forces $\text{err}_t = 1$ below $-c h(t-1)$. The accumulator cannot then
 128 leave $[-c h(t), c h(t)]$ by more than one round, and $|E_T| \leq c h(T) + 1$: Proposition 2's own bound,
 129 retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies
 130 $\tau \leq b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2's radius at
 131 $\mu_t = 1$.

132 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If $\hat{q}_t \equiv \hat{q}$, if
 133 $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t , then $\sigma^\pm = \tau^{*\pm}$ and **coverage is retained against every**
 134 **legal adapted adversary if and only if $\tau \leq \tau^{*+}$ and $\tau < \tau^{*-}$.** For a symmetric saturator that is
 135 one line: coverage if and only if $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when
 136 $\hat{q} < 0$. Every setting run here is in that sub-class, so the sharp form applies to all five. **What was**
 137 **printed before is the failure half.** τ^{*+} is the rule Figures 2 and 3 draw, and all 19 dead-band runs
 138 agree with it read against the adversary the harness plays. It is not a retention condition, and Table 2
 139 (App. A) separates the two with legal objects only. The supremum can sit where the dynamics never
 140 go: $|E_t| \leq (1 - \alpha)t$, so a saturator equal to the harness's inside $|x| \leq t^2$ and to $\pm 10b$ outside it
 141 satisfies (4), reads $\tau^{*+} = 19$, and fails at $\tau = 1.001$. A legal time-varying $\hat{q}$ splits $\sup_t$ from $\inf_t$:
 142 $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reads $\tau^{*+} = 2$ against $\sigma^+ = 1$. And that printed rule
 143 omits τ^{*-} entirely.

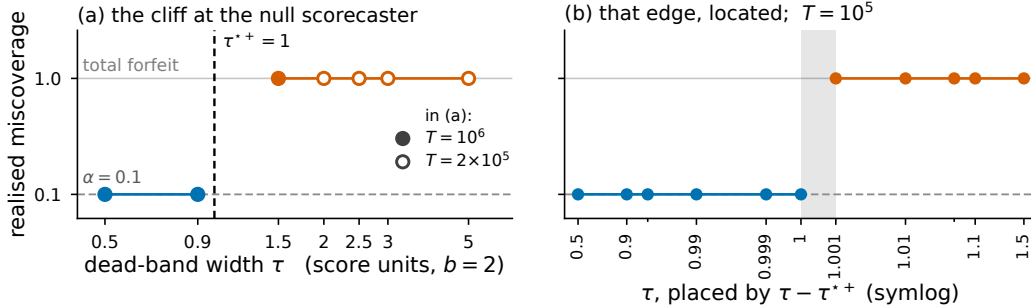


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , under the *specified* adversary, where the upward boundary τ^{*+} of (2) is 1. (a) The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$; blue covers, orange forfeits, and the dashed rule is τ^{*+} . (b) The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{*+}$ on a symlog axis whose linear zone is $|\tau - \tau^{*+}| \leq 10^{-3}$, which magnifies the neighbourhood of τ^{*+} : visual spacing here is not proportional to true distance in τ . Unlabelled minor ticks mark $\tau = 0.95$ and 1.05 . The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings. The same eleven widths under the *mirror* adversary, which locates τ^{*-} , are read off in Appendix A's primary-regime paragraph.

144 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric
 145 saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^\pm = A_t^\pm = \pm b$; the harness's
 146 $r_t = b \text{clip}(x/(c h(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

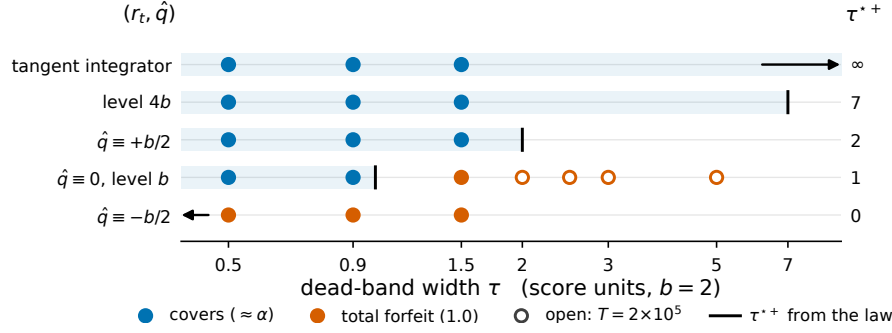


Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by τ^* , and the shared horizontal axis is τ . Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2 \times 10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \leq \tau^{*+}$. Two rules fall off the axis and are drawn as arrows: $\tau^{*+} = 0$ at $\hat{q} \equiv -b/2$, and τ^{*+} unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary τ^{*-} of (2) is not drawn here and differs from τ^{*+} at the two non-null scorecasters.

with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2's radius at $\mu_t = 1$, now as an if-and-only-if. **The necessity direction above assumes none of it:** $\tau > \tau^{*+}$, or $\tau \geq \tau^{*-}$, forfeits coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather than decorative: the unattained supremum and the varying- $\hat{q}$ split two paragraphs above are exactly what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they always do.

The mirror boundary is already measured, on this paper's own grid. The τ -cliff run's exact-threshold column under the strict indicator is the mirror adversary $s_t \equiv -b/2$ (one error process, $\text{err}_t = 1\{\hat{q}_t < -b/2\}$), and on the same eleven widths at the null scorecaster it locates $\tau^{*-} = 1$, closed on the failing side, exactly where (2) puts it; Appendix A prints the eleven readings. Both 1.000000 and 0.000000 are failures of the same order, $|E_T| = (1 - \alpha)T$ and αT , **so the failure criterion is $|E_T|$, not the miscoverage rate**; and once $\hat{q}$ varies the rate stops being a tell at all.

Both degenerate edges, and what they need. An unbounded supremum is not on its own enough for $\tau^{*+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$ past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(ch(t)))$, so the accumulator level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)ch(t)$ with $\kappa(\tau) = \arctan(b/2 + \tau - \hat{q})/\arctan(b)$, finite at every finite τ . Coverage is retained at every width, as measured; but $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2's own *constant* is forfeited past $\tau = b/2$ while coverage is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2's 14.8155 and against $\kappa(1.5)ch(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$ is at least $-b/2$ already. There $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t , and $\tilde{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

What that establishes is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$ the boundary is exactly Corollary 2's radius, crossing it costs coverage rather than the rate ($\tau = 0.9$ covers at 0.100012, $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900,000$, 60,747 times Proposition 2's bound at $T = 10^6$; Table 1), and (2) makes that an if-and-only-if against the whole legal adversary class rather than a pattern across five settings. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only, and now provably so: under the tangent integrator Corollary 2 permits $\tau \leq b/2$ while no finite width forfeits coverage against any legal adversary, because the reach is genuine at every level. At $\hat{q} \equiv +b/2$ the looseness goes the other way and disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary, the unsmoothed control included, though $\tau = 1.5$ covers there against the specified one. The

condition therefore binds at some legal settings and is slack at others, and that asymmetry is what makes “tight” a located claim rather than a global one: what is stated here is where the radius is sharp, and equally where it is not.

The constructive reading, and it is the corollary’s rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already. What Section 3 adds is that the band has two edges, both computable before deployment from the integrator’s guaranteed reach and the scorecaster’s range, that crossing either is not a graceful degradation, and that between $\sigma^\pm$ and $\tau^{*\pm}$, which separate only when $\hat{q}$ varies in t , neither direction is settled here.

The boundary read as a bet. The boundary is stated in $|E_T|$, which is the net gain of a Skeptic who pays α each round for err_t , so it has a betting reading. Impose, for that reading alone and on nothing else here, the null that $\text{err}_t \mid \mathcal{F}_{t-1}$ is Bernoulli(α), with $\mathcal{F}_{t-1}$ the history through round $t - 1$: past the edge an anytime-valid test of the deployed intervals’ calibration convicts the $\tau = 1.5$ arm on its fifth interval at the 5% level, and inside the edge the same test cannot reject at that level at any horizon. Appendix B gives the construction, the arithmetic and the two arms; the accumulator itself is signed and is neither an e-value nor an e-process, and no result above assumes that null or is certified by that test.

4 Where this sits

One quantity, four names. Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. Online conformal prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al., 2023, Thm 1, Prop. 2]. Forecast stability treats the movement as a functional, read out as post-processing [Godaheva et al., 2025, Eq. 3], against a priced decision to switch [Genov et al., 2026, §4.2], [Van Belle et al., 2026]. Verification work uses a matched-level design against real producers from 2012 [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning gives worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2].

The object that settles a deployed interval’s validity (the admissible set around the deployed threshold) sits in the first of them. Those four vocabularies are where this paper’s claim and its concessions have to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$ for a sublinear h , supported with a plot rather than derived. Verification’s matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013, Thm. 7].

The accumulator, read as a bet. Appendix B reads E_t as a game-theoretic capital process and builds a calibration test on top of the boundary above; it is not repeated here.

5 Limitations

The result above is narrow in several specific ways. They are listed here rather than left to be found, and the run-in bold marks where one concession ends and the next begins.

233 **The retention direction is proved with no gap only where the two boundaries meet:** a constant
234 scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run
235 here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{*+}$ open in neither direction (Section 3).
236 And $0.63/(1-w)$ is specific to the saturator’s *level*, which condition (4) bounds only from below.
237 **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$,
238 the certified gap $(c h(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was
239 not a tight certificate. **The dead-band widths are absolute, and the failure is total only against**
240 **the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75 b$. Under i.i.d. scores it
241 returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat.
242 The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly.

243 What follows concerns the evidence rather than the statement. **The harness was rebuilt against num-**
244 **bers already read**, which carries less weight than a rerun done without sight of them. **Placement B**
245 **is conceded, not tested:** the argument that it stays inside the admissible radius is Theorem 1’s, and no
246 arm run here exercises it. **The i.i.d. check above used one seed** (20260820, $T = 10^6$); **the primary,**
247 **adversarial harness uses none, and only its null-scorecaster edge is bracketed;** the others are
248 one-sided.

249 **What remains open is the band between the two boundaries.** For a genuinely time-varying $\hat{q}$,
250 coverage in $\sigma^+ < \tau \leq \tau^{*+}$ is neither proved nor falsified; Table 2’s power-of-two rows sit there, and
251 closing that band is future work, not a sweep this paper has already run.

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A Supporting measurements

The primary regime. Every number in Table 1 is measured under one deterministic adversary, which consumes no randomness and needs no seed. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$ against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and the distinction is measured rather than argued. Scoring an exact-threshold play as a *miscoverage* reproduces every quantity Table 1 reports: 44 of 44 arm-horizon cells identical, all eleven arms at all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every row, because the adversary’s signature move then covers rather than misses. Partial adjustment at $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place of 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading, and that 623.70 turns on the adversary’s ε , not the counting convention. **No row is convention-free with respect to the second,** which is why the adversary is specified rather than tie-broken. At $\tau = \tau^{*+}$ itself the two readings part: the strict indicator returns 0.000000 where scoring the exact play as a miscoverage returns 1.000000. **Neither of those is coverage.** They are $|E_T| = \alpha T$ and $(1 - \alpha)T$, failures of the same order and opposite sign. Under the strict indicator the exact-threshold play scores $\text{err}_t = \mathbf{1}\{\tilde{q}_t < -b/2\}$, which is also the indicator of the constant legal play $s_t \equiv -b/2$, so that column is Section 3’s *mirror adversary* rather than a second convention. It returns between 0.09992 and 0.09997 at $\tau = 0.5, 0.9, 0.95, 0.99$ and 0.999, and 0.000000 with $\max_t |E_t| = 10,000$ at $\tau = 1, 1.001, 1.01, 1.05, 1.1$ and 1.5. That locates $\tau^{*-} = 1$ on the same eleven widths Figure 2 uses, closed on the failing side. The convention decides which legal adversary realises the endpoint failure, not whether one occurs.

Excursions at $T = 10^4$, and where the excursion law holds. Partial adjustment excurses 6.30, 63.00 and 623.70 at $w = 0.9, 0.99$ and 0.999 . The last two are flat in T while Proposition 2's bound grows like $\log T$, which is why the ratio falls with the horizon: at $w = 0.999$ it is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. On the control and the four partial-adjustment arms the excursion fits $\max_t |E_t| \approx \max\{0.5 h(T) + 0.9, 0.63/(1 - w)\}$ at every horizon, most loosely at $w = 0.9$, the one arm whose two branches cross inside the horizon range. This is a fit to those five arms, not a derived bound. It does not extend to the other six: the growing time constants, the running mean and the dead bands all excure further than it predicts.

Realised miscoverage at the covering settings of Figure 3. At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.'s tangent integrator the same width returns 0.100001. Raising the saturator's level to $4b$ leaves coverage intact and cuts the $w = 0.999$ excursion from 623.70 to 120.60. That level is $\sup_x r_t(x)$, which is the first term of $\tau^{\pm}$ in (2), so what the change moves is the edge itself; the readout is unaltered and behaves no differently.

What survives of the measurement, and what its status now is. Outside the admissible window the transition is a step. Past τ^{*+} the threshold sits permanently below the reachable score range and $|E_t| = (1 - \alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; the wider bands $\tau = 2.0, 2.5, 3.0$ and 5.0 fail too at $T = 2 \times 10^5$ (Figure 2). **Here $\sup_x r_t(x) = b$ and $\hat{q} \equiv 0$, so $\tau^{*+} = \tau^{*-} = b/2$,** one point of the admissible class, and both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound against the specified adversary, and against the mirror one no width covers there, the unsmoothed control included. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the failure direction nothing; it moves the edge too (Figure 3) and shrinks the excursion, from 623.70 to 20.10 under the tangent integrator Angelopoulos et al. [2023] report using in all their experiments, both at $T = 10^6$. **The method's own integrator therefore sits inside the set at every width run here. The endpoint is where the two boundaries part.** The upper one is closed and the lower one is open, which the null scorecaster cannot show because τ^{*+} and τ^{*-} coincide there: four saturators with $\tau^{*+} < \tau^{*-}$ cover at $\tau = \tau^{*+}$ exactly under both adversaries, and at $\hat{q} \equiv +0.4$, where $\tau^{*-} = 0.6 < 1.4 = \tau^{*+}$, the width 0.599 covers under both while 0.6 fails under the mirror. At the baseline the two coincide at 1, so the primary regime covers there and the mirror adversary does not; Figure 2 and this appendix's primary-regime paragraph print both readings. **The failing arms have Sat. = 1.0000.** Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis: the threshold closing the loop is the integrator's output.

Table 2: Legal $(r_t, \hat{q})$ configurations on which the printed τ^* over-certifies, each under two legal adversaries. "Proved window" is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{*\pm}$ neither direction is claimed, and the rows built on a supremum the dynamics cannot reach or never attain sit there. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included. $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

$(r_t, \hat{q})$ configuration	τ	printed τ^*	proved window	$s_t = \hat{q}_t + \varepsilon$		$s_t \equiv -b/2$	
				miscov.	$\max_t E_t $	miscov.	$\max_t E_t $
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	0.000000	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	1.000000	90,000	0.000000	10,000
$A^+ = 4b, A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	0.000000	10,000
level $4b, \hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b, \hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	0.000000	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	0.000000	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	0.000000	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	0.110960	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0	2	$\emptyset$	0.100000	0.90	0.000000	10,000

Where the rate stops being a tell. At $\tau = 1.9$ with the power-of-two scorecaster the rate is 0.110960 against $\alpha = 0.1$ while $\max_t |E_t|$ is 84,745.70 at $T = 10^6$, so a reader watching only the rate would call that coverage.

413 B The accumulator as a bet, and a calibration test built on the boundary

414 **What is assumed here, and by whom.** Everything Section 3 proves is deterministic and pathwise.
 415 Proposition 2’s window $|E_T| \leq c h(T) + 1$, and the boundaries (2) that locate its edge, hold on
 416 every trajectory a legal adapted adversary can produce; no distributional model on the data appears in
 417 either statement, and neither is a statement about a null hypothesis. This section neither uses that
 418 nor weakens it. It runs the other way: it takes the error sequence a deployment actually produced as
 419 *data*, and tests a hypothesis about it. The hypothesis is imposed here, for the test alone, with $\mathcal{F}_{t-1}$
 420 the history through round $t - 1$ (scores, deployed thresholds and errors) against which $\hat{q}_t$ is already
 421 required to be predictable:

$$H_0 : \quad \text{err}_t \mid \mathcal{F}_{t-1} \sim \text{Bernoulli}(\alpha) \quad \text{for every } t,$$

422 and it says that the shipped intervals were conditionally calibrated at the target rate. H_0 is the object
 423 under test. It is not a premise of Proposition 2, of (2), or of any other claim in this paper; nothing
 424 above requires it, implies it, or is disturbed if it fails. When it fails, what loses validity is the test below
 425 and nothing else. We are explicit about this because the rest of the paper assumes no distribution at
 426 all, and a reader who meets a Bernoulli null on this page must not read it backwards into the pages
 427 before it.

428 **The accumulator is a capital process, and it is signed.** A Skeptic who each round pays α for the
 429 payoff err_t has cumulative net gain exactly E_t , so this paper’s accumulator is a game-theoretic capital
 430 process at unit stake [Shafer and Vovk, 2019, Ramdas et al., 2023]. It is not an e-value and not an
 431 e-process: it is signed, it goes negative on exactly the trajectories the mirror boundary τ^{*-} governs,
 432 and Proposition 2 bounds it pathwise by feedback rather than under a null by Ville’s inequality [Vovk
 433 and Wang, 2021]. Nothing below proposes E_t , or a transform of it, as a certificate for any claim this
 434 paper makes. What is formal is the standard transform, and it is invoked rather than invented.

435 **The test martingale, and why it is one.** Fix λ and set $M_0 = 1$,

$$M_t = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha)).$$

436 Each factor takes one of two values, $1 + \lambda(1 - \alpha)$ on a miss and $1 - \lambda\alpha$ on a cover, so $M_t \geq 0$
 437 for every path if and only if $\lambda \in [-1/(1 - \alpha), 1/\alpha]$, which for $\alpha = 0.1$ is $[-10/9, 10]$. Under H_0 ,
 438 $\mathbb{E}[\text{err}_t \mid \mathcal{F}_{t-1}] = \alpha$, so each factor is conditionally mean one and $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1}$: M_t is
 439 a nonnegative martingale started at 1, i.e. a test martingale for H_0 . Ville’s inequality then gives
 440 $\mathbb{P}_{H_0}(\sup_t M_t \geq 1/\eta) \leq \eta$, so “reject H_0 the first time $M_t \geq 1/\eta$ ” is an anytime-valid level- η test of
 441 the deployed intervals’ calibration, at any stopping time and with no horizon fixed in advance. That is
 442 a test of the deployment, not of Section 3.

443 **The boundary transfers exactly, because M_t sees only (t, E_t) .** Since $\text{err}_i \in \{0, 1\}$, the product
 444 collapses. Writing N_t for the miss count and using $N_t = \alpha t + E_t$,

$$\begin{aligned} \log M_t &= K(\lambda)t + L(\lambda)E_t, \\ K(\lambda) &= \alpha \log(1 + \lambda(1 - \alpha)) + (1 - \alpha) \log(1 - \lambda\alpha), \\ L(\lambda) &= \log \frac{1 + \lambda(1 - \alpha)}{1 - \lambda\alpha}. \end{aligned}$$

445 with $K(\lambda) \leq 0$, $K(\lambda) = 0$ only at $\lambda = 0$, and $\text{sign } L(\lambda) = \text{sign } \lambda$. So M_t depends on the path only
 446 through the horizon and the accumulator – the same statistic Section 3’s dichotomy is stated in – and
 447 the transfer is an identity rather than the first-order expansion $\log M_t = \lambda E_t + O(\lambda^2 t)$. It transfers
 448 as a dichotomy, not as a measured rate. Inside the retained band $|E_t| \leq c h(t) + 1$ is logarithmic, so
 449 $\log M_t \leq K(\lambda)t + |L(\lambda)|(c h(t) + 1)$ at every round. Since $K(\lambda) < 0$ the right side falls to $-\infty$
 450 linearly, so $M_T \rightarrow 0$ exponentially; and since it is continuous with a single interior maximum it has
 451 a finite supremum $S(\lambda)$ over all t , so $\sup_t M_t \leq e^{S(\lambda)}$ *uniformly in the horizon*. The test therefore
 452 never rejects at any level $\eta < e^{-S(\lambda)}$, however long the deployment runs. It is the uniform bound and
 453 not the limit that non-rejection needs, because Ville’s inequality prices the supremum. Past either
 454 boundary $|E_T|$ is linear in T , so some fixed λ of the sign of E_T makes $\log M_T$ grow linearly and the
 455 test rejects at any fixed level within $O(1)$ rounds. Both signs are needed, and that is the point of the
 456 two-sided form: the τ^{*+} failure drives E_T up and calls for $\lambda > 0$, the mirror failure at τ^{*-} drives it
 457 down and calls for $\lambda < 0$.

458 **Worked on this paper’s own frozen arms, at the null scorecaster.** Take the harness of Section 2
459 ($\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t + 2)$, $\hat{q} \equiv 0$), where $\tau^{*+} = \tau^{*-} = 1$, and the two dead
460 bands that bracket that edge in Table 1. *Past the edge*, $\tau = 1.5$ returns miscoverage 1.000000 and
461 $\max_t |E_t| = 900,000$ at $T = 10^6$; the run records $n_{\text{err}} = T$ at all four horizons, so the per-round path
462 is exactly $\text{err}_t = 1$ and $E_t = (1 - \alpha)t$, and no reconstruction is involved. A genuine pre-registered
463 deployment fixes λ , sign included, before the data arrive – by running both signs as one two-sided
464 test, or by a mixture over λ – and the single sign used per arm below is chosen after the fact, to
465 illustrate the identity on a trajectory already known to fail or to hold; it is not itself a claim of a
466 pre-registered test. At $\lambda = 1$ – a round number well inside $[-10/9, 10]$, not the aggressive endpoint
467 – the identity above gives $M_t = 1.9^t$, so at level $\eta = 0.05$, threshold $1/\eta = 20$: $M_4 = 13.03$ and
468 $M_5 = 24.76$. **The anytime-valid test rejects on the fifth deployed interval.** At $\eta = 0.001$ it rejects
469 on the eleventh ($M_{10} = 613.11$, $M_{11} = 1164.90$). *Inside the edge*, $\tau = 0.9$ covers at 0.100012
470 with $\max_t |E_t| = 13.20$ against Proposition 2’s 14.8155 at the same $T = 10^6$, and the same $\lambda = 1$
471 gives $\log M_t \leq -0.030639 t + 9.8632$ from the measured excursion alone: below $\log 20$ from round
472 225, below 0 from round 322, and $M_T \leq e^{-30,629}$ at $T = 10^6$. The measured excursion bounds
473 one run; the theorem bounds every one. This width sits inside $\sigma^\pm$, so the retention half applies and
474 $|E_t| \leq c h(t) + 1$ holds at every round, whence $S(1) = \sup_t (K(1)t + L(1)(\log(t+2) + 1)) = 2.448$,
475 attained at $t = 22$, and $M_t \leq 11.57$ at every round and every horizon. The test cannot reject at
476 $\eta = 0.05$ – not merely after round 225, but ever. Five rounds against never, on two arms of one
477 frozen grid. *The mirror direction*, on the same $\tau = 1.5$ configuration read under the strict indicator
478 against $s_t \equiv -b/2$: miscoverage 0.000000, $\text{err}_t = 0$ at every round, $E_t = -\alpha t$, and $\lambda = -1$ gives
479 $M_t = 1.1^t$, which crosses 20 at $t = 32$ ($M_{31} = 19.19$). The asymmetry is not a defect of the test: a
480 forced miss carries $\log 1.9$ nats per round against a forced cover’s $\log 1.1$, because $\alpha = 0.1$ makes
481 misses the rarer event under H_0 .

482 **What this adds, and what it does not.** It adds a diagnostic and a reading, not a result. The
483 reading is that E_t is the Skeptic’s capital and that Section 3 therefore locates, in a quantity a better
484 already cares about, the exact width at which a dead band turns a calibrated deployment into one that
485 an anytime-valid test convicts within a handful of rounds. The diagnostic is runnable online by a
486 practitioner who has the errors and not the proof. Neither statement is a claim about E_t ’s status: E_t
487 is not an e-process, M_t is a textbook transform imposed on top of the result rather than a part of it, and
488 this paper’s contribution is the tightness of someone else’s admissible radius, proved pathwise. M_t is
489 itself an e-value, and we say so: an e-value for H_0 alone, a hypothesis about one deployment’s error
490 sequence, imposed on this page and nowhere else in the paper. No result of this paper is an e-value
491 result or rests on one. Betting enters online conformal inference elsewhere as an optimiser rather
492 than as evidence [Podkopaev et al., 2024].